Aurick Anwar

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EDUCATION

McMaster University Hamilton, ON
Bachelor of Engineering, Engineering Physics *Expected 2029*

EXPERIENCE

Software Engineering Intern Sep 2026 – Present
Magic Hour AI (YC W24) *San Francisco, CA*

- Built a 2x image upscaler on FLUX.2 Klein in PyTorch, cutting memory use with 8-bit weights and CPU offloading so it runs on a single **16GB T4 GPU**.
- Built a prompt optimizer that tests 4 image models through the Magic Hour API, uses GPT-5.4 to score each result, and auto-rewrites prompts until they pass, covering **288 free image tools**.
- Parallelized image generation across **8 workers** with crash-safe resume, logging every result to Google Sheets.
- Built an async Python scraper that pulls text, pricing, images, and video from product pages using httpx and a stealth browser, then uses GPT-5.4 mini to turn it into structured brand data.

Software Engineer Sep 2026 – Present
McMaster Mars Rover Team *Hamilton, ON*

- Developed 10+ ROS2 nodes for perception, localization, and drive control on a Jetson AGX Orin with **<3 ms latency**.
- Ran SLAM within the ROS2 Nav2 stack to map terrain and localize the rover in real time for autonomous navigation.
- Integrated GPS/IMU localization into Nav2, enabling the rover to navigate to GPS waypoints **within 3 m**.

Software Engineer Jun 2026 – Present
McMaster Aerial Drone & Robotics Team *Hamilton, ON*

- Built a reproducible ArduPilot Copter SITL environment with Gazebo, enabling **15+ team members** to test autonomous missions without risking flight hardware.
- Automated MAVProxy waypoint missions over MAVLink, hitting **20/20 successful runs** with <1.2 m waypoint error.
- Injected GPS-loss failures to validate failsafes (**<2 s recovery**); built ROS2/pymavlink tools for 20 Hz telemetry.

Software Engineering Intern May 2026 – Aug 2026
HermesAI *Toronto, ON*

- Built the document upload and AI case-extraction flow, from React UI to REST APIs, for **1,000+ page** legal filings.
- Engineered a LangChain pipeline that parses, chunks, and extracts case law from long legal documents, cutting manual review time by **50%** for Canadian law firms.
- Cut document processing latency by **30%** by moving file parsing to asynchronous Node.js background jobs.

PROJECTS

MagnusBot | *LeRobot, RF-DETR, Roboflow, OpenCV, Next.js, Claude API* *Cal Hacks 13.0*

- Built a robot that plays live Chess.com games on a real board using **two SO-101 arms**, with custom analytic IK, per-piece grasps for all 6 piece types, and an arm-to-arm hand-off for cross-board moves.
- Trained an RF-DETR detector on 436 annotated images (**98.3% mAP**, 97.9% precision, 99.1% recall), mapping pieces to a 64-square grid from 4-corner calibration and running inference locally to cut latency.
- Polled Chess.com at 1 Hz, gating Claude Vision with a **0.1% pixel-diff** filter so unchanged frames skip the API.

NL Robot | *ROS2, Gazebo, FastAPI, Three.js, Docker, Claude API* *Hack the North 2026*

- Built a robot simulator where users upload a URDF and command it in plain English, cutting robot test setup time by **70%** compared with hand-written ROS2 commands.
- Turned commands into validated Claude API plans on a **20 Hz ROS loop** driving a diff-drive base and 4-DOF arm.
- Streamed robot state at **30 Hz** through a FastAPI bridge to a Three.js viewport, with raycast 360° LiDAR in Docker.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, HTML/CSS, MATLAB
ML & Computer Vision: PyTorch, TensorFlow, Scikit-learn, OpenCV, YOLOv11, LangChain, MediaPipe
Robotics & Simulation: ROS2, Nav2, LeRobot, RViz, Gazebo, SLAM, MuJoCo, ArduPilot, MAVLink
Embedded & Hardware: NVIDIA Jetson, Arduino, Raspberry Pi, ESP32
Web & Tools: React, Node.js, FastAPI, Flask, REST APIs, Linux, Git, Docker